

Vulnerability assessment

OpenVAS/Greenbone

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May 13, 2026

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Chapter 1

System Discovery

In this chapter we will describe how to perform a System Discovery using OpenVAS. There must be clarified that this type of scan is not a Vulnerability Scan, it's just a detection of the targets of a system, their open ports, the services running on them and its whole topology.

1.1 Get Debian VM IP address

Before performing the System Discovery, we need to know the IP address of the Debian VM, since we need to specify it as Greenbone default gateway to reach the docker network where the vulnboxes are located. Open the Debian VM, which credentials are:

- Username: root
- Password: root

Then go to the folder `/network-security/project/docker-vulnet/` and run the command `bash hardcore-setup.sh` to burn down the firewall and build up some rules which allow the Debian machine to redirect the traffic to the docker network. This bash script flush all the iptables rules and then add the minimum number of them to allow the Debian machine to redirect the traffic to the docker network, with proper NATting and forwarding rules.

```
Debian GNU/Linux 13 debian tty1
debian login: root
Password:
Linux debian 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@debian:~# cd network-security/project/docker-vulnet/
root@debian:~/network-security/project/docker-vulnet# bash hardcore-setup.sh
[+] down 10/10
+ Container vulnbox6 Removed
+ Container vulnbox1 Removed
+ Container vulnbox2 Removed
+ Container vulnbox3 Removed
+ Container vulnbox4 Removed
+ Container vulnbox5 Removed
+ Network docker-vulnet_net1 Removed
+ Network docker-vulnet_net2 Removed
+ Network docker-vulnet_net3 Removed
+ Network docker-vulnet_net0 Removed
[+] up 10/10
```

Figure 1.1: Debian VM - Initial setup

After that, run the command `ip addr | grep 192` to get the IP address of the Debian VM.

```
root@debian:~# ip a | grep 192
    inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
root@debian:~# _
```

Figure 1.2: Debian VM - IP address

1.2 Configure Greenbone Gateway

Before performing the System Discovery, we need to configure the Greenbone Gateway to reach the docker network where the vulnboxes are located. Open the OpenVAS VM, which credentials are:

- Username: admin
- Password: admin

Go to **Setup** > **Network** > **Global Gateway** and change the default gateway according to the IP address of the Debian VM, which we got in the previous section. Then save the configuration.



Figure 1.3: Greenbone VM - Gateway configuration

1.3 Greenbone Web Interface

Open the *About* section in the main menu of the Greenbone VM.

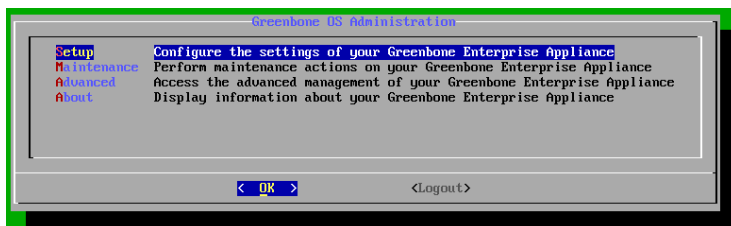


Figure 1.4: Greenbone VM - Main section

Check out the IP address of the VM.

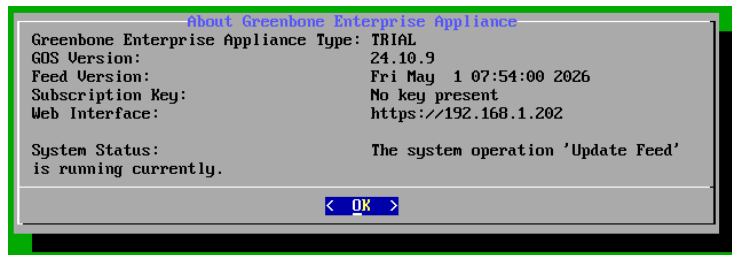


Figure 1.5: Greenbone VM - About section

Open a web browser and type the URL for the web interface in the search bar.

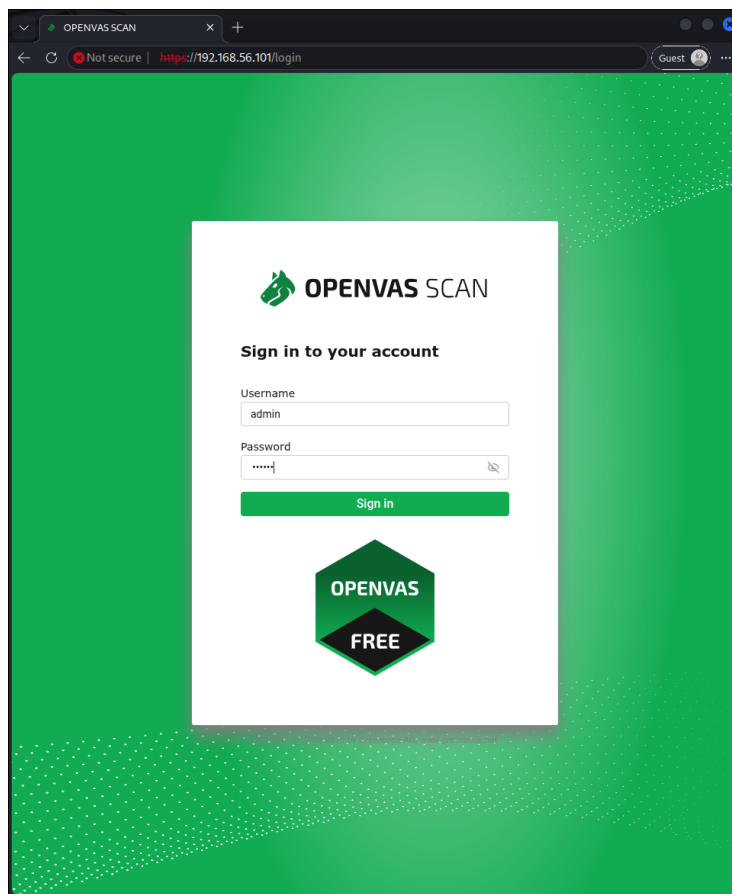


Figure 1.6: Greenbone Web Interface - Login page

Log in with the credentials:

- Username: admin

- Password: admin

You will see the main page of the web interface.

1.4 System Scan

Now go to the *Scans* section and click on *Tasks*.

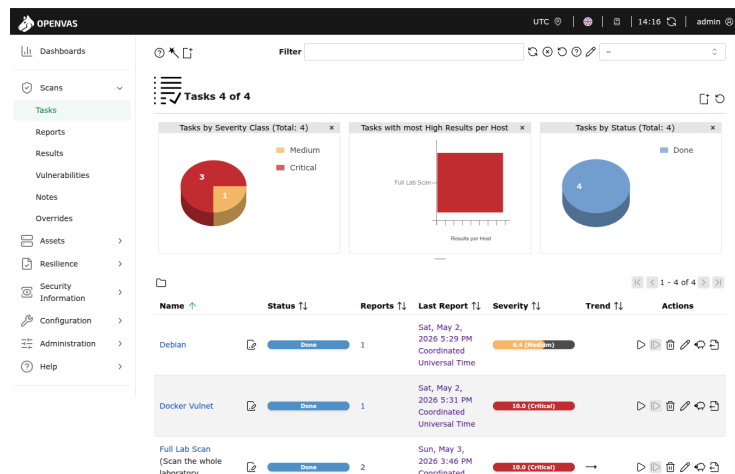


Figure 1.7: Greenbone Web Interface - Main page

Modify the *Docker Vulnet* task. Edit it by choosing *System Discovery* as the scan configuration, then save the task.

Edit Task Docker Vulnet

Scan targets

Vulnet Target

Add results to Assets

☒ Yes ☐ No

Apply Overrides

☒ Yes ☐ No

Min QoD

70

Alterable Task

☐ Yes ☐ No

Auto Delete Reports

☒ Do not automatically delete reports

☐ Automatically delete oldest reports but always keep newest

5

reports

Scanner

OpenVAS Default

Scan Config

System Discovery

Order for target hosts

Sequential

Maximum concurrently executed NVTs per host

7

Maximum concurrently scanned hosts

7

Cancel

Save

Figure 1.8: Greenbone Web Interface - Task modification

Now we can launch the task pressing the *Play* button.



Figure 1.9: Greenbone Web Interface - Task launch



Figure 1.10: Greenbone Web Interface - Task progress

After about 10 minutes, the task will be completed and we can see the results. To see the results, click on the task and then on the *Report* tab. In the upper right corner, click on the *Remove filters* button to see all the results. This is required because by default the web interface shows only the results with a severity score, but since the *System Discovery* scan configuration doesn't detect vulnerabilities, all the results have a severity score of 0, so we need to remove the filters to see them all.

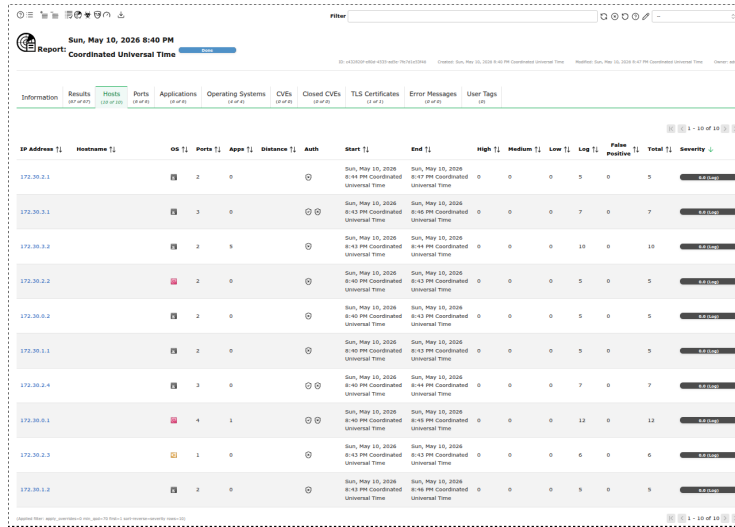


Figure 1.11: Greenbone Web Interface - Remove filters

Now we can see all the results of the scan which comprehends:

- The detected hosts in the given IP range
- The open ports on the hosts
- The services running on the open ports
- The topology of the system

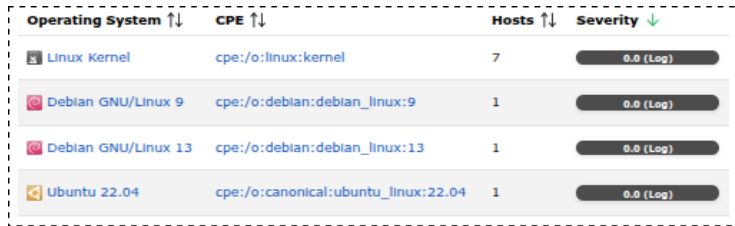
Even if the scanned system only has 6 vulnboxes, as we can see, OpenVAS detected a lot more hosts, since in the OpenVAS view, each IP address corresponds to a host, and in the scanned system some vulnboxes have more than one IP address, therefore OpenVAS detected all of them as distinct hosts.



The screenshot shows the Greenbone Web Interface (GWI) displaying scan results for a target IP address 172.30.2.1. The interface includes a top navigation bar with tabs for Information, Results, Hosts, Ports, Applications, Operating Systems, CVEs, Closed CVEs, TLS Certificates, Error Messages, and User Tags. The main content area shows a table of scan results with columns for IP Address, Hostname, OS, Ports, Apps, Distance, Auth, Start, End, High, Medium, Low, Log, False Positive, Total, and Severity. The table lists 10 scan results for the target IP, showing various ports and applications detected.

IP Address	Hostname	OS	Ports	Apps	Distance	Auth	Start	End	High	Medium	Low	Log	False Positive	Total	Severity
172.30.2.1			2	0			Sun, May 10, 2026 8:14 PM Coordinated Universal Time	Sun, May 10, 2026 8:17 PM Coordinated Universal Time	0	0	0	5	0	5	0.0 (Log)
172.30.3.1			3	0			Sun, May 10, 2026 8:12 PM Coordinated Universal Time	Sun, May 10, 2026 8:16 PM Coordinated Universal Time	0	0	0	7	0	7	0.0 (Log)
172.30.3.2			2	0			Sun, May 10, 2026 8:13 PM Coordinated Universal Time	Sun, May 10, 2026 8:14 PM Coordinated Universal Time	0	0	0	10	0	10	0.0 (Log)
172.30.2.2			2	0			Sun, May 10, 2026 8:14 PM Coordinated Universal Time	Sun, May 10, 2026 8:15 PM Coordinated Universal Time	0	0	0	5	0	5	0.0 (Log)
172.30.2.2			2	0			Sun, May 10, 2026 8:14 PM Coordinated Universal Time	Sun, May 10, 2026 8:15 PM Coordinated Universal Time	0	0	0	5	0	5	0.0 (Log)
172.30.3.1			2	0			Sun, May 10, 2026 8:12 PM Coordinated Universal Time	Sun, May 10, 2026 8:16 PM Coordinated Universal Time	0	0	0	5	0	5	0.0 (Log)
172.30.2.4			2	0			Sun, May 10, 2026 8:14 PM Coordinated Universal Time	Sun, May 10, 2026 8:14 PM Coordinated Universal Time	0	0	0	7	0	7	0.0 (Log)
172.30.3.1			4	1			Sun, May 10, 2026 8:13 PM Coordinated Universal Time	Sun, May 10, 2026 8:13 PM Coordinated Universal Time	0	0	0	12	0	12	0.0 (Log)
172.30.2.3			1	0			Sun, May 10, 2026 8:13 PM Coordinated Universal Time	Sun, May 10, 2026 8:13 PM Coordinated Universal Time	0	0	0	6	0	6	0.0 (Log)
172.30.1.2			2	0			Sun, May 10, 2026 8:14 PM Coordinated Universal Time	Sun, May 10, 2026 8:14 PM Coordinated Universal Time	0	0	0	5	0	5	0.0 (Log)

Figure 1.12: Greenbone Web Interface - Scan results



The screenshot shows the Greenbone Web Interface (GWI) displaying operating system results for a target IP address 172.30.2.1. The interface includes a top navigation bar with tabs for Operating System, CPE, Hosts, and Severity. The main content area shows a table of operating system results with columns for Operating System, CPE, Hosts, and Severity. The table lists 4 operating systems detected: Linux Kernel, Debian GNU/Linux 9, Debian GNU/Linux 13, and Ubuntu 22.04.

Operating System	CPE	Hosts	Severity
Linux Kernel	cpe:/o:linux:kernel	7	0.0 (Log)
Debian GNU/Linux 9	cpe:/o:debian:debian_linux:9	1	0.0 (Log)
Debian GNU/Linux 13	cpe:/o:debian:debian_linux:13	1	0.0 (Log)
Ubuntu 22.04	cpe:/o:canonical:ubuntu_linux:22.04	1	0.0 (Log)

Figure 1.13: Greenbone Web Interface - Operating Systems

The results of this scan can be also be seen in the *Assests;Hosts* section of the web interface, where we can see all the detected hosts and their open ports and services.

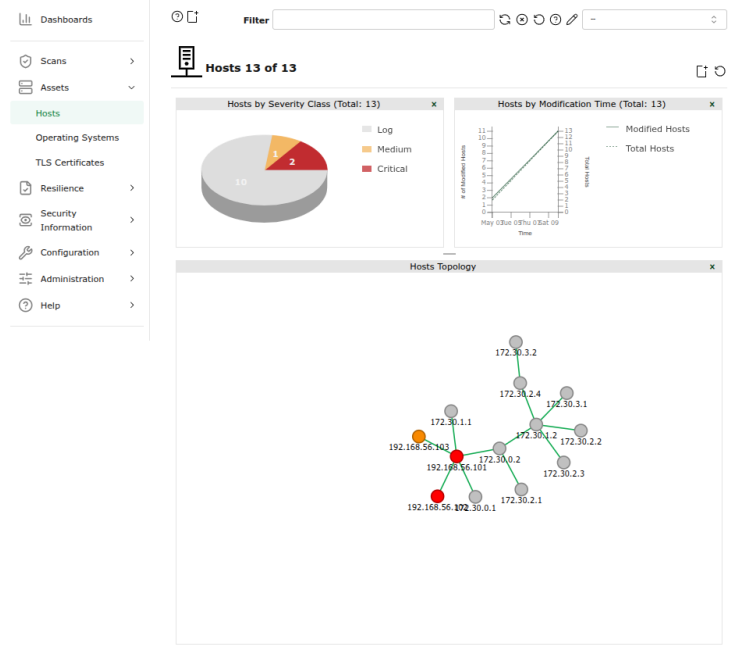


Figure 1.14: Greenbone Web Interface - Hosts section

We can here check if the detected topology of the system is correct, by comparing it with the one we designed.

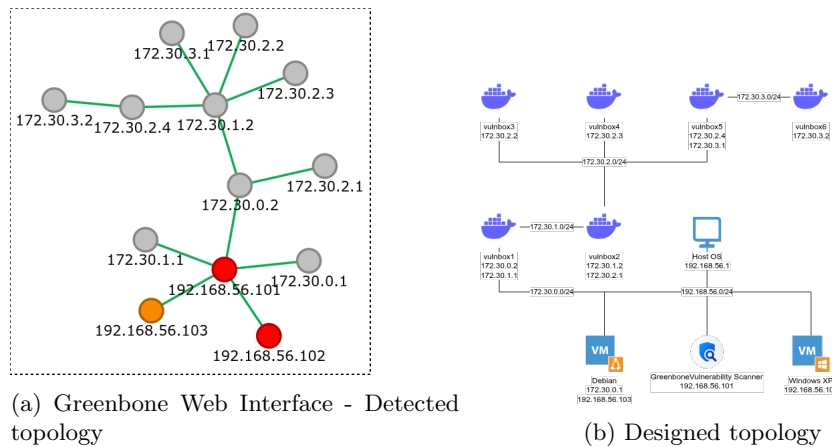


Figure 1.15: Greenbone Web Interface - Detected topology vs Designed topology

As we can see, the detected topology is correct, even if there are more hosts than vulnboxes, the overall topology is connected in the right way.

Chapter 2

AI Usage Declaration

This report was prepared with the assistance of AI tools for drafting and editing support. The structure and the core contents of the report were designed by the author. All final content was reviewed and approved by the author.